

AAYUSH PANDEY

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Education

VIT Bhopal University

Aug 2023 – Aug 2027

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.24

Madhya Pradesh

Experience

DRDO – Winter Trainee | [Github](#) | Nov 2025 – Jan 2026

(Python, React, Node.js, Express.js, PostgreSQL, Prisma ORM, Ollama (Llama 3), REST APIs)

- Engineered a full-stack OSINT platform, automating entity extraction from documents and metadata to reduce manual data processing time by 75% for investigative tasks.
- Built an iterative investigation pipeline to extract entities from documents and integrated Ollama (Llama 3) for keyword extraction and report generation.
- Crafted a normalized PostgreSQL database schema for 6+ core entities, improving data integrity and enabling efficient relationship mapping.
- Created 5+ custom OSINT automation scripts to gather and enrich intelligence from public web sources.

DRDO – Summer Trainee | [Github](#) | Apr 2026 – Jun 2026

(Python, TensorFlow, TensorFlow Lite, MediaPipe, face-api.js, React, Node.js, Express.js, PostgreSQL)

- Developed a real-time ASL-to-text platform recognizing ~99% gestures accurately across 26 hand signs, reducing communication latency to under 100 ms.
- Implemented face recognition using face-api.js and TensorFlow-based hand gesture classification model using MediaPipe integrated into a React application.
- Engineered a backend with 6 RESTful API endpoints and an administrative dashboard with JWT authentication.
- Structured a PostgreSQL database capable of managing 1000+ facial descriptors and user/admin records for real-time recognition.

Projects

MindConnect | [Github](#) | [Live Demo](#)

(TypeScript, React, Node.js, Express.js, Prisma ORM, PostgreSQL (Neon), JWT Authentication, Vercel)

- Architected a full-stack mental health platform with 10+ REST API endpoints across admin, auth, booking, and doctor management modules.
- Modelled a PostgreSQL schema reducing query redundancy and improving data consistency across 5 core modules using Prisma ORM for type-safe database access.
- Implemented JWT-based authentication, securing 100% of user routes and enabling role-based access control for admin operations.
- Deployed on Vercel with Neon PostgreSQL, using a monorepo architecture with shared TypeScript types across client and server.

MediScan | [Github](#) |

(TypeScript, React, Node.js, Express.js, Prisma ORM, PostgreSQL (Neon), Vercel)

- Built a cross-platform desktop application to process DICOM images for automated fat and muscle tissue segmentation, improving workflow efficiency compared with conventional manual methods.
- Created a custom CNN-based image processing model to improve tissue detection accuracy and reliability (30px edge filter, 50px minimum region threshold).
- Developed an interactive DICOM editing tool enabling region-of-interest selection and export with preserved metadata and auto-generated SOP Instance UIDs.
- Modeled a SQLite-based storage system to manage 10+ clinical metrics.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, SQL

Frameworks & Runtime: React, Node.js, Express.js, Tailwind CSS, Electron.js, RESTful APIs

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Oracle SQL

ML & Data: TensorFlow, TensorFlow Lite, Keras, OpenCV, MediaPipe, face-api.js, NumPy, scikit-image, pydicom

Developer Tools: VS Code, Vite, Linux (Ubuntu, Arch, Kali), Bash

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Vercel, Git, GitHub